

gray matter
white matter
Parkinson's disease
Basal Ganglia
Substantia Nigra
Broca's Area
Wernicke's Area
Tan
Frontal lobe
Fluent Aphasia
Expressive Aphasia

Nernst potential
Ionic concentration gradient
Membrane voltage
diffusive/chemical/entropic vs. electrical force/work
Na⁺/K⁺ ATPase
leak channels
resting potential
reversal potential
membrane resistance (units)
membrane capacitance (units)
membrane current (units)
membrane time constant (units, typical range)
inhibitory/excitatory post-synaptic potential
membrane integration (role of R, C, τ)
role of ionic charge, concentration differences in reversal potentials
typical chloride reversal potential 50-65 mV
typical potassium reversal potential -100--80 mV
typical sodium reversal potential -80--60 mV
typical resting potential -75--60 mV

Action potential
threshold
positive feedback
upswing, depolarisation
repolarisation
hyperpolarisation
relative refractory period
absolute refractory period
which ions flow, in which directions, in each step

Hodgkin Huxley (**memorise mathematical model** with physiological interpretations)
voltage-gated ionic conductances
voltage gates
maximal conductances
voltage-gated sodium m gate (role in spiking)
voltage-gated sodium h gate (role in spiking)
voltage-gated potassium n gate (role in spiking)
voltage clamp
patch clamp (cell attached)
whole cell
stable, scalar, linear, first-order ODEs
steady state ($t \rightarrow \infty$) solutions
time constants
exponential solution ansatz
Forward Euler integration
Forward exponential Euler integration
two compartment models
axial conductance, currents
there are some HH deep thought questions!

LIF (**memorise mathematical model** with physiological interpretations)
find its threshold
adding a refractory period
derive relationship between spiking frequency/period and applied current
A-level differential calculus, critical thought

soma
dendrite
axon
synapse
axonal terminal / axonal bouton / presynapse
dendritic spine / postsynapse
action potential propagation (no PDE math)
steps in synaptic transmission
calcium influx in presynapse
readily releasable pool
vesicle fusion
neurotransmitter release and diffusion
neurotransmitter binding
ligand vs voltage-gated ion channels
metabotropic vs ionotropic neurotransmitter receptors
short-term depression/facilitation (**memorise mathematical model** with physiological interpretations)

Dale's principle

Modulators: Serotonin, histamine, dopamine, adrenaline/norepinephrine, (acetylcholine?)

AMPA receptors **ionotropic**, glutamate, non-selective cation channel, reversal near 0 mV

NMDA receptors **metabotropic**, glutamate, Ca^{2+} channel, coincidence detector using Mg^{2+} block), reversal near 0 mV <- Nitrous oxide, Dextrorphan, Ketamine

GABA_A receptor is **ionotropic chloride** channel (-60 to -80 mV)

GABA_B receptor is **metabotropic** channel (usually activating a **potassium** current) (-80 to -100 mV) <- Benzodiazepines

Muscarinic acetylcholine receptor (mAChR) metabotropic (various effects)

Nicotinic acetylcholine receptor (nAChR) non-selective cation channel, reversal near 0 mV <- Nicotine

Hebbian learning

Oja's rule

Finding fixed points and checking stability in nonlinear ODEs in one variable

LTP, LTD, STDP

gradient descent parameter optimisation, loss function

A-level differential calculus for converting loss functions to gradient descent rules

hippocampus

entorhinal cortex (EC)

hippocampal formation

areas DG, CA3, CA1

granule cells

sparse k-winner-take all pattern separation in DG (function?)

auto-associative (attractor, Hopfield-like) recurrent network in CA3 (function?)

CA1 memory readout, place cells

EC input-output, grid cells

Hopfield networks (**memorise mathematical model** with physiological interpretations)

What is an attractor network?

Hopfield energy function

Calculating Hopfield weights

Calculating Hopfield state evolution

cerebellum
granule cells
Purkinje cells
climbing fibre error
binary perceptron McCulloch-Pitts (linear threshold) "neuron" (memorise
mathematical model with physiological interpretations)
mathematical definition: inputs, weights, Heaviside step threshold, outputs
delta learning rule
Evaluating a perceptron/McCulloch-Pitts unit's output for given input
Updating a perceptron/McCulloch-Pitts unit's weights

photoreceptors
bipolar cells
retinal ganglion cells
center-surround tuning
opponent channels
Gabor function (Gaussian * sinusoid)
simple cells (mostly V1 layer 4, some 6)
complex cells (V1 layer 2/3 and higher)
phase invariance
tuning curve
receptive field
what pathway
where pathway
cortical hierarchy
retinotopy
Olshausen and Field's sparse over-complete basis/dictionary model of simple cells
Wilson-Cowan neural fields
Rate coding (temporal and population senses)
Phase coding (hippocampal theta phase precession)
Timing code (inter-aural time delays)